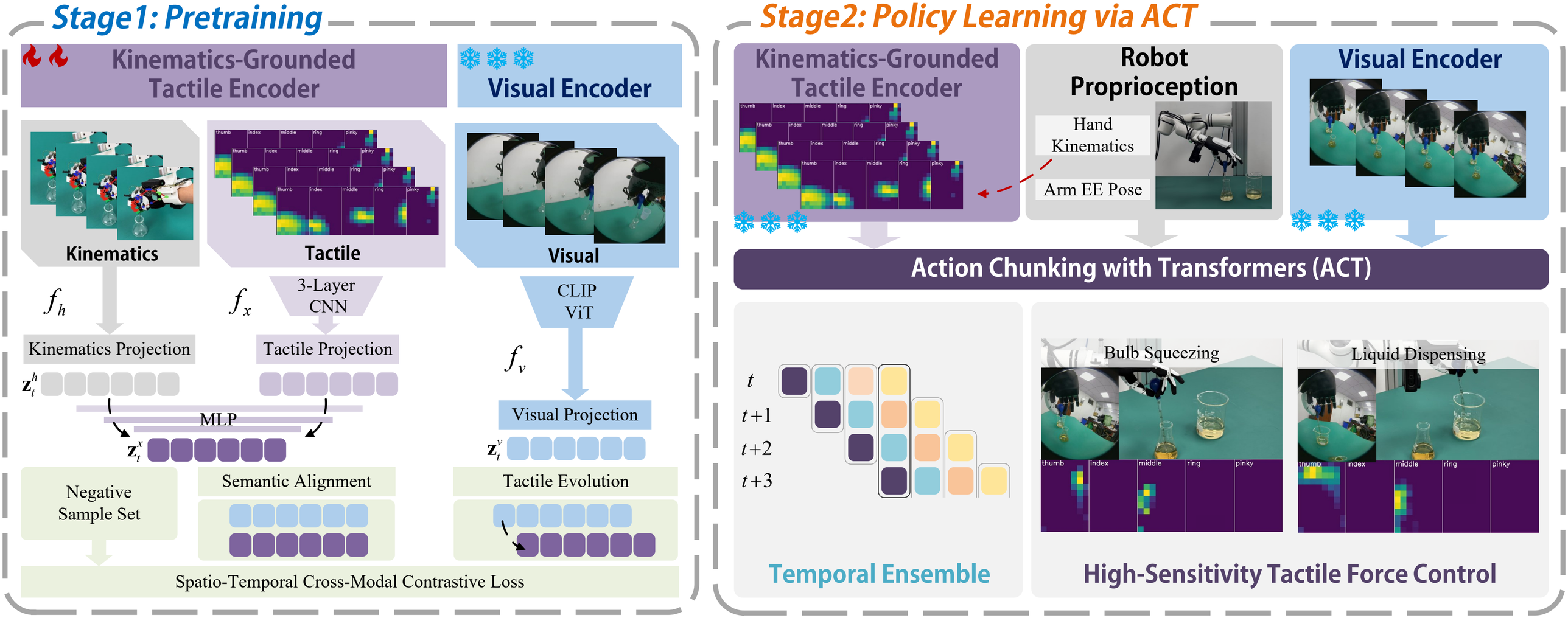
Stage1: Pretraining
Kinematics-Grounded Tactile Encoder
Visual Encoder
Kinematics
Tactile
Visual
f_h
f_x
3-Layer CNN
CLIP ViT
f_v
Kinematics Projection
Tactile Projection
Visual Projection
$\mathbf{z}_t^h$
MLP
$\mathbf{z}_t^x$
$\mathbf{z}_t^v$
Negative Sample Set
Semantic Alignment
Tactile Evolution
Spatio-Temporal Cross-Modal Contrastive Loss
Stage2: Policy Learning via ACT
Kinematics-Grounded Tactile Encoder
Robot Proprioception
Hand Kinematics
Arm EE Pose
Visual Encoder
Action Chunking with Transformers (ACT)
t
$t+1$
$t+2$
$t+3$
Temporal Ensemble
Bulb Squeezing
Liquid Dispensing
thumb
index
middle
ring
pinky
High-Sensitivity Tactile Force Control